

Decision Problem. Function with a one-bit output : "Yes" or "No".

Set of possible inputs to a decision problem

Set of finite length strings over some fixed finite alphabet.

Alphabet - finite set - denoted by Σ

Ex. $\{0, 1, 2, \dots, 9\}$ - decimal numbers,

$\{0, 1\}$ -

$a, b \in \Sigma$

Strings over Σ - finite length sequence of elements of Σ .

Ex. $\Sigma = \{a, b\}$ $abbaa$ - string of length 5

x, y, z - denote strings.

length of a string x - $|x|$ - number of symbols in x .

Unique string of length 0 - null string.

$\in [\text{epsilon}]$.

$$|\epsilon| = 0.$$

$a \in \Sigma$, a^n - string of a 's of length n .

$$a^0 = \epsilon \quad a^{n+1} = a^n a$$

Set of all strings over an alphabet Σ - denoted by Σ^* .

Ex.

$$\{a, b\}^* = \{\epsilon, a, b, aa, ab, ba, bb, aaa, \dots\}.$$

$$\{a\}^* = \{\epsilon, a, aa, aaa, \dots\}.$$

$$= \{a^n \mid n \geq 0\}$$

$$\emptyset^* = \{\epsilon\}$$

String and sets are not the same.

$$\{a, b\} = \{b, a\} \quad / \quad ab \neq ba$$

$$\{a, a, b\} = \{a, b\} \quad / \quad aab \neq ab$$

$\emptyset$ - empty set

ϵ - null string.

$\{\epsilon\}$ - Set with one element - null string.

Operations on Strings

Concatenation. 2 strings x & y and creates a new string xy

Note. $xy \neq yx$ are in general different

- Concatenation is associative $(xy)z = x(yz)$
- null string ϵ is identity for concatenation

$$\epsilon x = x\epsilon = x$$

- $|xy| = |x| + |y|$

Let $x \in \Sigma^*$

x^n - string consisting of n copies of x .

Example: $(abb)^3 = abbabbabb$.

Formally x^n can be defined inductively.

- $x^0 = \epsilon$.
- $x^{n+1} = x^n x$

Prefix of a string.

Prefix of x is a string y s.t there exists z with $x = yz$.

- Null string is a prefix of every string.
- Every string is a prefix of itself.

Operations on sets. : subsets of Σ^*

A, B - sets of strings.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\},$$

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Complement $\bar{A} = \{x \in \Sigma^* \mid x \notin A\}.$

Set concatenation

$$AB = \{xy \mid x \in A \text{ and } y \in B\}.$$

$$\begin{array}{c} \downarrow \quad \nearrow \\ \{a, ab\} \quad \{b, ba\} = \{ab, aba, abb, abba\} \end{array}$$

Note: In general AB and BA are different.

A^n - inductively

$$A^0 = \{\epsilon\} \quad A^{n+1} = AA^n$$

$$\{ab, aab\}^0 = \{\epsilon\}$$

$$\{ab, aab\}^1 = \{ab, aab\}.$$

$$\{ab, aab\}^2 = \{abab, abaab, aabab, aabaab\}.$$

$$\{a, b\}^n = \{x \in \{a, b\}^* \mid |x| = n\}.$$

A^* - Union of all finite powers of the set A

Σ^* - Set of all finite strings over Σ .

$$A^* = \bigcup_{n \geq 0} A^n = A^0 \cup A^1 \cup A^2 \cup \dots$$

$$A^* = \{x_1 x_2 \dots x_n \mid n \geq 0 \text{ and } x_i \in A, 1 \leq i \leq n\}$$

ϵ is in A^* for any A - Since n can be 0.

$$A^+ = AA^* = \bigcup_{n \geq 1} A^n$$

Properties of Set operations

$$\left. \begin{aligned} (A \cup B) \cup C &= A \cup (B \cup C) \\ (A \cap B) \cap C &= A \cap (B \cap C) \\ (AB)C &= A(BC) \end{aligned} \right\} \begin{array}{l} \text{Commutative} \\ \text{Associative} \end{array}$$

Concatenation is not commutative

$\{\epsilon\}$ - Identity for concatenation.

$$\{\epsilon\}A = A\{\epsilon\} = A$$

$$A\phi = \phi A = \phi$$

ϕ is an annihilator for set concatenation.

* satisfies the following properties:

$$A^* A^* = A^*$$

$$A^{**} = A^*$$

$$\phi^* = \{\epsilon\}$$

Deterministic Finite Automata (DFA)

$$M = (Q, \Sigma, \delta, s, F)$$

Q - finite set of states.

Σ - input alphabet (a finite set)

$s \in Q$: Start state of M

$F \subseteq Q$: Set of Final/accept states of M .

$\delta : Q \times \Sigma \rightarrow Q$ is a transition function.

M is in some state q and reads input "a"

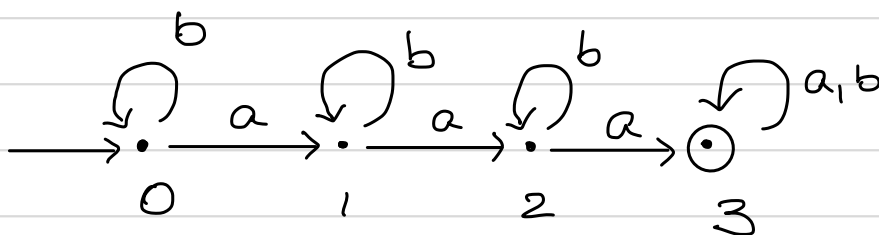
$\delta(q, a) = q'$; M moves to state q'

Example:

$$Q = \{0, 1, 2, 3\}; \Sigma = \{a, b\}; s = 0; F = \{3\}$$

Transition Function :

$$\begin{array}{l|l} \delta(0, a) = 1 & \delta(2, a) = \delta(3, a) = 3 \\ \delta(1, a) = 2 & \delta(q, b) = q \quad q \in \{0, 1, 2, 3\}. \end{array}$$



$\rightarrow \bullet$: Start state.

$\odot$: Final State

a is accepted by M $\delta(s, a) \in F$

DFA $M = (Q, \Sigma, s, \delta, F)$

Input $x \in \Sigma^*$. M runs on input x .

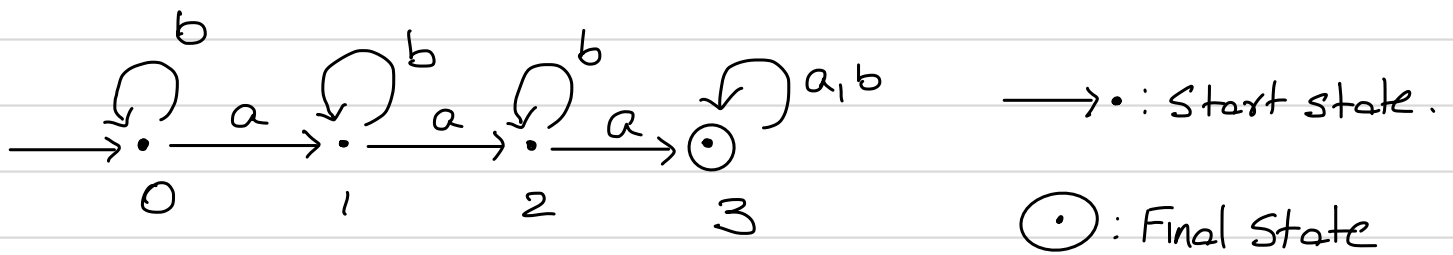
How does M process the input string x

- M starts in the initial state s .
- Scan x from left to right reading one symbol at a time
- Suppose the first symbol in x is $a \in \Sigma$.
- M reads a and changes its state to $q = \delta(s, a)$
- Repeat the process for each subsequent symbols.

How does M decide whether to accept or reject x

- After reading all symbols in x , M is in some state $q \in Q$.
- if $q \in F$ then M accepts x
- if $q \notin F$ then M rejects x

Example



Input $x = aabba$

M accepts x since $3 \in F$.

Input $y = abbab$

After reading y , M is in State 2 and $2 \notin F$.

So M rejects y .

$$L(M) = \{ x \in \{a,b\}^* \mid x \text{ contains at least } 3 \text{ a's} \}.$$

DFA $M = (Q, \Sigma, s, \delta, F)$ where. $s: Q \times \Sigma \rightarrow Q$

We can "lift" s as a transition function over strings.

$$\hat{s}: Q \times \Sigma^* \rightarrow Q.$$

Defining $\hat{s}$ inductively.

$$\hat{s}(q, \epsilon) = q$$

$$\hat{s}(q, xa) = s(\underbrace{\hat{s}(q, x), a})$$

well defined by induction

Claim 1. We can show that $\hat{s}(q, a) = s(q, a)$

$$\begin{aligned}\hat{s}(q, a) &= \hat{s}(q, \epsilon a) \quad [\text{Since } a = \epsilon a] \\ &= s(\hat{s}(q, \epsilon), a) \\ &= s(q, a)\end{aligned}$$

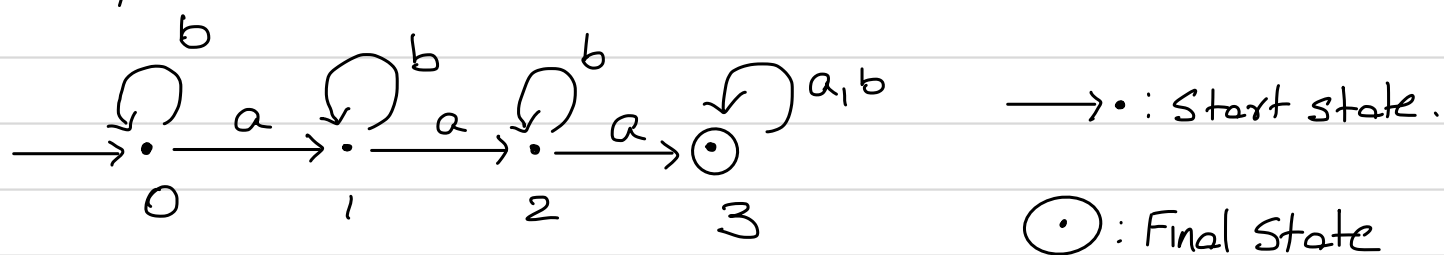
x is accepted by M if $\hat{s}(s, x) \in F$
 x is rejected by M if $\hat{s}(s, x) \notin F$

Language of the DFA M .

$$L(M) = \{x \in \Sigma^* \mid \hat{s}(s, x) \in F\}$$

$A \subseteq \Sigma^*$ is regular if $A = L(M)$ for some DFA M .

Example:



Input $x = a a b b a$

M accept x since $\hat{\delta}(0, x) = 3$ and $3 \in F$.

Input $y = a b b a b$

M rejects y since $\hat{\delta}(0, y) = 2$ and $2 \notin F$

$L(M) = \{ x \in \{a, b\}^* \mid x \text{ contains at least three } a\text{'s} \}$.

$A = \{x \in \{a,b\}^* \mid x \text{ contains a substring of three consecutive a's}\}$.

$b a a b b \underline{a a a} b b \in A$

$b a a b b a b b \notin A$

Question. Is A regular?

To show that A is regular it suffices to construct a DFA M such that $L(M) = A$.

